

Defense Speech — 15 minutes

Intelligent Similarity Search System for the Gemba Walk Assistant

Mohamed Mortadha Agoubi · iObeya · 2025/2026

How to use this document

The speech is written as **short lines, one idea per line**. Do not memorise it word for word. Memorise the **order of the ideas**. The words will come.

Symbol	Meaning
▶ CLICK	press the right arrow — the slide advances one step
(pause)	stop talking for one second. It is not silence, it is punctuation.
	a line you can cut if you are running late
	your safety net — if you lose your words, say this one sentence and move on

Total spoken text: ~1 480 words ≈ 12 minutes 30.

That leaves you 2 to 3 minutes for hesitation, breathing, and improvisation.

The three rules that save you

1. **Speak slowly.** Slower than feels natural. Your English is judged on *clarity*, never on speed.
2. **If you lose a word, do not go back.** Use the  line for that slide and move to the next one. Nobody in the room knows what you planned to say.
3. **Every slide has one job.** If you remember only the job, you can improvise the rest. The jobs are: *context, pain, idea, proof, trust*.

Words to practise out loud

Word	Say it like	Note
Gemba	GEM -ba	hard G, like "get"
Obeya	o- BAY -ya	
iObeya	i-o- BAY -ya	
semantic	se- MAN -tic	
meaning	MEE -ning	you will say this word ~10 times
retrieve / retrieval	ri- TREEV / ri- TREE -val	
threshold	THRESH -old	say "thresh-old", two parts
precision	pre- SI -zhun	
recall	ri- CALL	
partition	par- TI -shun	
milliseconds	MI -li-seconds	
archive	AR -kaive	

If a word feels dangerous, you are allowed to replace it. "retrieve" → "find". "the archive" → "the old issues". "semantic" → "by meaning". Simple English is professional English.

THE SPEECH

Slide 1 — Cover · ~45 s

(Stand still. Look at the jury. Smile. Then start.)

Good morning everyone.

Thank you for being here today.

My name is Mohamed Mortadha Agoubi.

I did my End-of-Studies Project at iObeya.

(pause)

My subject is an Intelligent Similarity Search System
for the Gemba Walk Assistant.

In simple words:

I taught a factory tool to **remember**.

To remember every problem the company has already solved.

(pause)

Let me show you why this matters.



I taught a factory tool to remember the problems the company already solved.

Slide 2 — Agenda · ~35 s · 3 clicks

Here is where we are going together.

► **CLICK**

First, the context — where the data comes from.

Then the problem. And it is a serious one.

Then the solution — and I will show it working, on the real system.

Then what makes it different from a normal search.

And finally, the results.

► **CLICK**

One idea holds all of this together.

Every problem a company solves is an **asset**.

Today, that asset is filed away — and forgotten.



Context, problem, solution, difference, results.

Slide 3 — Lean · ~40 s · 3 clicks

Let me start with Lean.

Lean is a way of working.

Create more value, with fewer resources.

Remove waste. And solve problems where they happen.

► **CLICK**

These are the five principles of Lean. 🕒

I will not go through them one by one. 🕒

► **CLICK**

I only need three words from this world.

Gemba — it means "the real place". The shop floor.

Gemba Walk — you go there, and you see the problem with your own eyes.

And **continuous improvement** — every problem you solve becomes a standard.

(pause)

Please keep that last one in mind. It is the heart of my project.



Gemba means the real place. A Gemba Walk is going to see the problem yourself.

Slide 4 — Obeya → iObeya · ~45 s · 4 clicks

So where does iObeya come from?

"Obeya" is a Japanese word. It means "the big room".

At Toyota, they put the whole project on the walls of one room.

Everyone comes in. Everyone sees the same thing. They decide together.

(point at the photos on the left)

For many years, this was paper. On walls. In one building.

► **CLICK**

iObeya made it digital.

The same rituals. The same boards. But no walls.

► **CLICK**

Today it is used by Sanofi, Air France, Renault, Airbus, Thales, Volvo.

These are factories where one problem costs real money.



iObeya is the digital version of the Toyota "big room".

Slide 5 — The Gemba Walk Assistant · ~55 s · 4 clicks

Inside iObeya, there is one application for the shop floor.

It is called the **Gemba Walk Assistant**.

A manager walks the floor. He sees a problem. He writes it down.

But not as free text.

► **CLICK**

He fills a standard Lean form — the **5W2H**.

What happened. Who. Where. When. How. How many. And why it matters.

► **CLICK**

Then the team works on the problem.

They contain it. They find the root cause. They apply a solution.

And they write the new standard.

► **CLICK**

(slow down here — this is the key sentence of the first part)

And here is the important part.

When the problem is solved, the issue **stays** on the board.

With its root cause. Its solution. And the actions that worked.

(pause)

So the company is already storing all the answers.



Every solved problem stays on the board, with its solution attached.

Slide 6 — The problem · ~60 s · 3 clicks

(Let the video run 3 or 4 seconds before you speak. Let them watch.)

So if the answers are already there...

where is the problem?

(pause — let the video breathe)

Here it is.

A new problem is reported today.

Somewhere in that archive, the very same problem was already solved.

Maybe two years ago. Maybe by another team, in another building.

► **CLICK**

But nobody remembers it.

And nobody can find it.

► **CLICK**

So the team starts again. From zero.

They investigate. They test. They lose weeks.

And the company pays twice for the same answer.



The answer already exists. Nobody can find it.

Slide 7 — What it costs · ~40 s · 3 clicks

Why can they not find it?

Because the only tools are human memory, and scrolling.

And a board can hold thousands of cards.

► **CLICK**

So four things happen.

Time is lost.

The same problem is solved twice.

Good solutions stay buried.

And nobody sees which problems keep coming back.

► **CLICK**

(slow, clear — this is your best sentence of the problem part)

The company is not short of knowledge.

It is short of a way to **reach** it.

 **The knowledge is there. The way to reach it is not.**

Slide 8 — Search by meaning · ~55 s · 4 clicks

Now you may think: just use the search bar.

I tried. It does not work.

Look at these two issues.

Today, the operator writes:

"Scratches on machined parts after the tool change."

In the archive, we have:

"Surface defects on milled components."

► **CLICK**

It is the same problem.

But there is not one word in common.

A keyword search finds **nothing**.

► **CLICK**

So the system must compare **meaning**, not words.

To do that, we turn each text into a list of numbers. We call it a vector.

Texts that mean the same thing end up close to each other.

(point at the animation)

► **CLICK**

Text becomes meaning. Meaning is compared. And we get a ranked list.

That is the whole idea of this project.

 **Same problem, different words. So we compare meaning, not words.**

Slide 9 — The solution · ~2 min 20 · 5 clicks

(This is the centre of your defense. Do not rush. Let each clip play.)

So this is what I built.

Three steps. **Capture. Find. Reuse.**

Let me show you the real system.

► **CLICK** *(clip 1 — let it play, then speak over it)*

Step one: **Capture.**

Nothing changes for the operator.

He fills the 5W2H exactly like he does today.

No new habit. No new tool to learn.

► **CLICK** *(clip 2)*

Step two: **Find.**

One click.

The system takes this new issue,
and compares it with the whole history.

Not by keywords — by meaning, and field by field.

Here, more than two hundred issues, searched in a few milliseconds.

It gives back the closest ones, ranked.

(point at the score bars)

And look — it also shows **why** each one matched.

► **CLICK** *(clip 3)*

Step three: **Reuse.**

This is the part I care about the most.

The system does not only say "this looks similar".

It gives back the root causes that were already found.

The solutions that already worked.

And a ready action plan.

(pause)

And every single line says which issue it comes from.

The manager can create these actions on his own board, in one click.

► **CLICK**

Capture. Find. Reuse.

Inside the tool the teams already use every day.



Three steps: capture the problem, find what happened before, reuse what worked.

Slide 10 — Built on 5W2H · ~1 min 10 · 5 clicks

Now, what makes this different from a normal search engine?

A normal engine takes the whole issue and treats it as one block of text.

I do not do that.

► CLICK

An industrial problem has a **structure**. So I compare it field by field.

And every field does not have the same importance.

(point at the bars)

"What happened" carries a third of the score.

"Who saw it" carries almost nothing —

because the same failure seen by two operators is still **one** failure.

► CLICK

And two fields are not text at all.

A date is compared as a date.

A quantity is compared as a ratio.

► CLICK

The result is this:

the system can explain **why** two problems match.

Field by field. Not just a number. 🕒

► CLICK

And I found this by testing. 🕒

My first version put all the fields into one single block. 🕒

The best documented issues were ranked **last**. 🕒

So I changed the design. 🕒

 **A problem is not a paragraph. It has structure, and I use that structure.**

Slide 11 — Grounded AI · ~1 min · 5 clicks

There is AI in this project.

So the honest question is: can we trust it?

Here is my answer.

► **CLICK**

The action plan is **not written by the AI**.

It is built by the code, from the issues we found.

Every proposed action already exists, in a real past issue.

► **CLICK**

The AI is allowed to summarise. To rephrase. To explain.

► **CLICK**

It is **not** allowed to invent an action.

Not allowed to decide what should be done.

Not allowed to invent evidence.

► **CLICK**

And it works with no paid API, and no internet connection.

(slow — last sentence)

The AI writes. The expert decides.



The AI writes the text. It never invents an action. The expert decides.

Slide 12 — Results · ~1 min 20 · 4 clicks

Does it actually work? I measured it.

One thousand issues indexed.

Two hundred and seven test questions.

Precision around sixty percent. Recall around sixty-five.

► **CLICK**

But this next number is the one I am proud of.

I took the same two hundred and seven questions,
and I wrote them again with **different words**. Industry synonyms.

Like a real user would.

(pause — then slowly)

A keyword search lost **fourteen percent** of its quality.

My system lost **zero point two percent**.

► **CLICK**

Because in real life, people never reuse the exact same words.

That gap is the whole point of this project.

And it is fast: twenty-five to forty milliseconds to search the whole history.

► **CLICK**

Now I want to be honest with you.

My test data is mostly generated, and it comes from a single room.

So these numbers show that the system **works**.

They do not yet prove how it behaves in a real factory, at full scale.

That is the next step.



When the words change, keyword search loses 14%. My system loses 0.2%.

Slide 13 — Testing changed the architecture · ~50 s · 5 clicks

One last story. Thirty seconds. It is my favourite one.

To make the search faster, I added a standard index. Everyone does this.

► **CLICK**

Then I tested with five hundred issues.

And obvious matches came back **missing**.

► **CLICK**

So I investigated.

The index cuts the data into one hundred parts.

And it was searching only **one** of them.

One percent of the history.

► **CLICK**

(slow, one word at a time)

Nothing crashed. No error. No warning.

The answers were simply **wrong**.

► **CLICK**

I removed the index.

Now it searches everything, exactly, in twenty-five to forty milliseconds.

(pause)

I tell you this because it is the most important thing I learned this year.

A system can be broken and silent at the same time.

Only testing finds that.



My index was hiding 99% of the data. Nothing crashed — the answers were just wrong.

Slide 14 — Conclusion · ~50 s · 3 clicks

To conclude.

Three things make this system what it is.

It is **semantic** — it searches by meaning, not by words.

It is **domain-aware** — it is built on the Lean 5W2H, not on generic text.

And it is **grounded** — it returns real solutions, from real issues.

► **CLICK**

What comes next: a real production corpus, validation by Lean experts, and hybrid search to also catch machine codes and part numbers.

► **CLICK**

(stop. look at the jury. say it slowly.)

And this is the sentence I would like you to remember:

Every problem a company solves
should only have to be solved **once**.



Semantic, domain-aware, grounded. Solve it once.

Slide 15 — Thank you · ~20 s

Thank you for your attention.

I will be happy to answer your questions.

(Then stop talking. Stand straight. Wait.)

Q&A — prepare these ten

Short answers. Never defend — **explain**. If you do not know, say *"I did not test that. My honest answer is..."* A jury respects that far more than an invented answer.

1. Why not simply use ChatGPT for everything?

Three reasons. Cost — it must work without a paid API. Privacy — factory data should not leave the company. And most important: a language model **invents**. My action plan must come from real past issues. So retrieval does the work, and the AI only writes the text.

2. Your test data is mostly generated. Are the results meaningful?

For the absolute numbers — not yet, and I say so in the report. For the comparison — yes. Both engines were tested on exactly the same data, with the same questions. So the difference between minus fourteen percent and minus zero point two is a fair comparison.

3. Sixty percent precision is not very high.

It is a strict measure. I count a result as correct only if it shares **both** the same root cause **and** the same solution. A genuinely useful issue that was fixed a different way counts as an error in my numbers. So the real usefulness is higher than sixty percent.

4. Why is the threshold at 0.45?

It is a recall-oriented default: I prefer to show a little more, because the interface shows the score and the reason for each match, so the user can judge. I measured the alternative — around 0.55 to 0.60 gives more precision and less recall. It is a setting, not code, so each deployment can choose.

5. Does it replace the expert?

No. It never decides. It proposes what worked before, and says where it comes from. The manager stays responsible. That was a design rule from the beginning.

6. What if the problem is completely new?

Then nothing passes the threshold, and it returns an empty list. That is a deliberate choice. I prefer to return nothing than to force a bad match and lose the user's trust.

7. Will it still be fast with one million issues?

Today it compares the query with every issue, so the time grows with the data. It is comfortable up to roughly ten to a hundred thousand issues. Above that, we bring back an approximate index — but this time we measure its recall first. That is exactly the lesson from my index problem.

8. Why did you choose the weights by hand instead of learning them?

Because my corpus is partly generated. If I learned the weights from it, I would learn the habits of my data generator, not the judgement of a Lean expert. So the weights come from Lean practice, they are configuration, and the report gives the procedure to re-tune them on real data.

9. Does it work in French, or other languages?

The current model is trained mainly on English. There is a multilingual model with the same size and the same vector dimension, so switching is a configuration change, not a rewrite. I did not measure it, so I cannot promise the quality yet.

10. Does any factory data leave the company?

No. The default model runs locally, offline, with no API key. That was a requirement from iObeya, and the whole core feature respects it.

One-page cue card

Print this page alone. Put it on the table. Glance, do not read.

#	Slide	The one thing to say
1	Cover	I taught a factory tool to remember .
2	Agenda	Context · problem · solution · difference · results
3	Lean	Gemba = the real place. Go and see.
4	Obeya	iObeya = the digital Toyota "big room". Sanofi, Airbus, Renault...
5	Gemba Assistant	5W2H. Solved issues stay on the board, with the solution.
6	Problem 🎬	The answer exists. Nobody can find it. They pay twice.
7	Cost	Not short of knowledge — short of a way to reach it.
8	Meaning	Same problem, no shared words → compare meaning .
9	Solution 🎬🎬🎬	Capture → Find → Reuse. Let the clips play.
10	5W2H	A problem is not a paragraph. What = 1/3. Who ≈ 0.
11	Grounded AI	AI writes. Code builds the plan. Expert decides.
12	Results	Words change: keyword -14% , mine -0.2% .
13	Testing	Index hid 99%. Nothing crashed. Answers were wrong.
14	Conclusion	Semantic · domain-aware · grounded. Solve it once.
15	Thank you	Thank you. Questions?

Numbers you must not get wrong: 1 000 issues · 207 questions · 60% / 65% ·
-14% vs -0.2% · 25–40 ms · 100 partitions, 1 searched.

Good luck. You know this project better than anyone in the room.